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IMAGE SENSING DEVICE FOR SIDE-VIEWING ENDOSCOPE

Abstract

An image sensing device for a side-view endoscope includes a substrate having a first surface, wherein a wire connection member, an image module connection member and a light source module connection member are disposed on the first surface, and wherein the wire connection member is electrically connected with the image module connection member and the light source module connection member; an image module electrically connected with the image module connection member and configured to capture external images along a lateral-side direction of a distal tip; and a light source module electrically connected with the light source module connection member, configured to project a light source along the lateral-side direction of the distal tip. By disposing the image module and the light source module on the same surface of the substrate, the device can be applied to side-view endoscopes and has the competitive advantage of miniaturization.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present invention relates to an endoscope, particularly to an image sensing device for a side-viewing endoscope.

2. Description of the Prior Art

[0002] Endoscopes have been widely used in industry and medicine. In industry, endoscopes may be used to inspect a narrow space, such as interiors of channels or complicated mechanical structures, without dismantling these structures, whereby to assist in inspection, improve product yield, and prolong the service lives of machines.

[0003] In medicine, endoscopes are often used in minimum invasive surgery and health examination, whereby to decrease invasiveness and avoid tissue damage. The smaller the endoscope, the better the feeling of the patient, and the fast the wound repair.

[0004] However, the size of endoscopes is hard to reduce at present, especially the front-viewing endoscopes.

SUMMARY OF THE INVENTION

[0005] One objective of the present invention is to provide an image sensing device for a small-size side-viewing endoscope.

[0006] The preset invention proposes an image sensing device for a side-viewing endoscope, which is disposed in a distal tip of the endoscope and comprises a substrate, a connection wire, an image module, and at least one light source module. The substrate includes a first surface and a second surface opposite to the first surface. The substrate further includes a wire connection member, an image module connection member and at least one light source module connection member, which are all arranged on the first surface. The wire connection member is electrically connected with the image module connection member. The wire connection member is also electrically connected with the light source module connection member. The image module captures external images along a lateral-side direction, which is between two ends of the distal tip. The light source module is electrically connected with the light source module connection member, projecting light along the lateral-side direction of the distal tip.

[0007] It is learned from the above description: the present invention improves the endoscope from the conventional front-viewing type into a side-viewing type. In other words, the image module captures images along the lateral-side direction of the distal tip. Further, the present invention provides a design able to minimize the overall size: disposing the connection wire, the image module and the light source module all on the first surface of the substrate. Thereby, the size of an endoscope is effectively reduced, and the objective of the present invention is achieved.

[0008] The objective, technologies, features and advantages of the present invention will become apparent from the following description in conjunction with the accompanying drawings wherein certain embodiments of the present invention are set forth by way of illustration and example.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The foregoing conceptions and their accompanying advantages of this invention will

become more readily appreciated after being better understood by referring to the following detailed description, in conjunction with the accompanying drawings, wherein

[0010] FIG. **1** is a perspective view schematically an appearance of an image sensing device for a side-viewing endoscope according to one embodiment of the present invention;

[0011] FIG. **2** is a perspective view schematically an image sensing device for a side-viewing endoscope according to one embodiment of the present invention;

[0012] FIG. **3** is a perspective view schematically a substrate of an image sensing device for a side-viewing endoscope according to one embodiment of the present invention;

[0013] FIG. **4** is a perspective view schematically substrates, image modules and light source modules of an image sensing device for a side-viewing endoscope according to one embodiment of the present invention;

[0014] FIG. **5** is a perspective view schematically a light source module of an image sensing device for a side-viewing endoscope according to one embodiment of the present invention;

[0015] FIG. **6** is a perspective view schematically a light source module of an image sensing device for a side-viewing endoscope according to another embodiment of the present invention;

[0016] FIG. **7** is a perspective view schematically a light shield of an image sensing device for a side-viewing endoscope according to another embodiment of the present invention;

[0017] FIG. **8** is a perspective view schematically substrates, image modules, light source modules and light shields of an image sensing device for a side-viewing endoscope according to one embodiment of the present invention;

[0018] FIG. **9** is a perspective view schematically a substrate, an image module, light source modules, a light shield, and a sensor of an image sensing device for a side-viewing endoscope according to one embodiment of the present invention;

[0019] FIG. **10** is a perspective view schematically a substrate, an image module, light source modules, a light shield, a sensor and an optical protection layer of an image sensing device for a side-viewing endoscope according to one embodiment of the present invention;

[0020] FIG. **11** is a perspective view schematically an image sensing device for a side-viewing endoscope according to another embodiment of the present invention; and

[0021] FIG. **12** is a perspective view schematically an image sensing device for a side-viewing endoscope according to yet another embodiment of the present invention.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0022] The embodiments of the present invention will be further demonstrated in details hereinafter in cooperation with the corresponding drawings. In the drawings and the specification, the same numerals represent the same or the like elements as much as possible. For simplicity and convenient labelling, the shapes and thicknesses of the elements may be exaggerated in the drawings. It is easily understood: the elements belonging to the conventional technologies and well known by the persons skilled in the art may be not particularly depicted in the drawings or described in the specification. Various modifications and variations made by the persons skilled in the art according to the contents of the present invention are to be included by the scope of the present invention.

[0023] Refer to FIG. **1** and FIG. **2**. In some embodiments of the present invention, the image sensing device for a side-viewing endoscope is disposed in a distal tip **900** of the endoscope and comprises a substrate **100**, a connection wire **200**, an image module **300**, and at least one light source module **400**.

[0024] Refer to FIG. **3**. The substrate **100** includes a first surface **110** and a second surface **120** opposite to the first surface **110**. In the embodiments, the substrate **100** is a ceramic substrate, a printed circuit board (PCB), or a flexible printed circuit (FPC). In the embodiments, the substrate **100** functions as a carrier. The substrate **100** further includes a wire connection member **130**, an image module connection member **140**, and at least one light source module connection member **150**, which are all arranged on the first surface **110**. The wire connection member **130** is electrically

connected with the image module connection member **140**. The wire connection member **130** is also electrically connected with the light source module connection member **150**. In fabricating the image sensing devices, the substrates **100**, which have not be separated, are arranged in array, whereby to simplify the fabrication process and favor mass production. In addition to a rectangular shape, the substrate **100** may alternatively have another shape to meet different requirements.

[0025] In some embodiments, the wire connection member **130**, the image module connection member **140** and the at least one light source module connection member **150** are arranged along a first direction **D1**. The first direction **D1** is a direction along a first end **111** of the first surface **110** and a second end **112** of the first surface **110**. Thus, the wire connection member **130**, the image module connection member **140** and the at least one light source module connection member **150** are compactly arranged along the first direction **D1**. Thereby, the image sensing device and the distal tip **900**, which are at the front end of the endoscope, may be diametrically minimized in a lateral-side directional **D2** of the distal tip **900**. It should be noted: the lateral-side directional **D2** of the distal tip **900** is a direction parallel to the lateral side of the distal tip **900** and substantially vertical to the first direction **D1**, as shown in FIG. 2.

[0026] In some embodiments, the present invention may use a single light source module connection member **150**. Along the first direction **D1**, the light source module connection member **150** is disposed on the first surface **110** and at one side of the image module connection member **140**, which is corresponding to the first end **111**; the wire connection member **130** is disposed on the first surface **110** and at one side of the image module connection member **140**, which is corresponding to the second end **112**. In other embodiments, along the first direction **D1**, the light source module connection member **150** is disposed on the first surface **110** and at one side of the image module connection member **140**, which is corresponding to the second end **112**; the wire connection member **130** is disposed on the first surface **110** and at one side of the image module connection member **140**, which is corresponding to the first end **111**.

[0027] In some embodiments, the present invention may use two light source module connection members **150**. Along the first direction **D1**, the light source module connection members **150** are disposed on the first surface **110** and respectively at one side of the image module connection member **140**, which is corresponding to the first end **111** and at one side of the image module connection member **140**, which is corresponding to the second end **112**. The wire connection member **130** is disposed on the first surface **110** between the light source module connection member **150** and the first end **111**. Alternatively, the wire connection member **130** is disposed on the first surface **110** between the light source module connection member **150** and the second end **112**.

[0028] Refer to FIG. 1 and FIG. 4. The image module **300** normally includes an optical lens module. The image module **300** captures external images along the lateral-side direction **D2** of the distal tip **900**. The image module **300** is joined and fixed to the image module connection member **140** in a high-temperature joining method. The image module **300** is electrically connected with and installed on the image module connection member **140** through the electrodes on the bottom thereof (BGA (Ball Grid Array) herein).

[0029] In some embodiments, the light source module connection members **150** are disposed on the first surface **110** and at either side or both sides of the image module connection member **140**. The substrate **100** may have one, two, three or more light source module connection members **150**. The light source module **400** may project light along the lateral-side direction **D2** of the distal tip **900**.

[0030] Refer to FIG. 3 and FIG. 4. In some embodiments, the image sensing device includes two light source module connection members **150**. Two pairs of the light source module connection members **150** and the light source modules **400** are respectively disposed on left and right sides of the image module **300**. The light source module **400** and the corresponding light source module connection member **150** are joined and secured in a high-temperature joining method, whereby the light source module **400** is electrically connected with and installed on the light source module

connection member **150**.

[0031] The light source module **400** may include a fluorescent powder layer **410** and a light-emitting element for emitting white light. Alternatively, the light source module **400** may be a light-emitting diode. In some embodiments, the light source module **400** may have light-emitting elements able to emit respectively lights with different wavelength; the wire connection member **130** has control wires connected with the light-emitting elements; the control wires receive control signals to turn on/off the light-emitting elements respectively emitting lights of different wavelengths.

[0032] The light source module **400** may emit light of the same wavelength and the same color temperature or emit lights of different wavelengths and different color temperatures, such as white light or infrared light. The lights of different wavelengths and different color temperatures may be turned on simultaneously or separately to meet different applications of the image sensing device.

[0033] Refer to FIG. 5. In some embodiments, the image sensing device further includes a carrier **500**, which is disposed between the light source module **400** and the light source module connection member **150**. The carrier **500** works to let the light source module **400** have different heights. The relative height of the light source module **400** and the image module **300** may be accurately controlled through using the carriers **500** having different heights. The size of the carrier **500** may be obtained via simulation and test.

[0034] Refer to FIG. 6. In some embodiments, the image sensing device further includes an intermediate carrier **510**, which is disposed between the carrier **500** and the light source module **400**.

[0035] Refer to FIG. 7. In some embodiments, the image sensing device further includes a light shield **600**, which is disposed between the light source module **400** and the image module **300**. In some embodiments, the light shield **600** is disposed around the light source module **400**. In some embodiments, the light shield **600** is disposed around the image module **300**. Refer to FIG. 8. In some embodiments, the light shield **600** surrounds the light source module **400** and the image module **300**.

[0036] It should be mentioned particularly: the light shield **600** may be made of a light-absorptive material, a light-reflective material, a light-scattering material, or a light-diffusive material. The abovementioned materials can prevent stray light from entering the image module **300** as well as homogenize the light source. The light shield **600** neither shades the top regions of the light source module **400** and the image module **300** nor shades the surrounding connection members. For example, the light shield **600** does not shade the wire connection member **130** lest the light generation efficiency, the image capture efficiency and the electric connection efficiency be affected.

[0037] The material of the light shield **600** may be filled into the space between the image module **300** and the light source module **400** and the surrounding regions to prevent from stray light and protect the image module **300** and the light source module **400**.

[0038] Refer to FIG. 9. In some embodiments, the image sensing device further includes a sensor **700**, such as a pressure sensor or a thermosensitive sensor. The sensor **700** is electrically connected with a sensor connection member **160** on the first surface **110** of the substrate **100**, as shown in FIG. 3. The sensor connection member **160** is electrically connected with the wire connection member **130**. The sensor **700** is used to detect the temperature or pressure of the image sensing device so as to provide more information for the operator.

[0039] Refer to FIG. 10. In some embodiments, an optical protection layer **800** is formed on the surface of the light source module **400**. The optical protection layer **800** covers the light source modules **400**, but it does not cover the image module **300**.

[0040] Refer to FIG. 11. In some embodiments, the connection wire **200** is electrically connected with the wire connection member **130** in a high-temperature soldering method. The connection wire **200** may be disposed and extended out according to requirement, whereby the size of the

image sensing device may be minimized. Because the connection wire **200** is directly soldered to the wire connection member **130** without using an adapter board or another adapter material, the image quality is guaranteed. After the connection wire **200** is soldered, a wire protection layer **210** may be formed on an area corresponding to the wire connection member **130**. The wire protection layer **210** and the light shield **600** may be designed to be an integral component and formed simultaneously in a single operation after the connection of the connection wire **200**. Thus, the connection points between the connection wire **200** and the sensor **700**, the image module **300** and the light source module **400** are also protected. Alternatively, the wire protection layer **210** and the light shield **600** may be fabricated separately, as shown in FIG. 2, wherein the wire protection layer **210** may be made of a UV-cured resin or a thermoset resin.

[0041] Refer to FIG. 12. In some embodiments, the image sensing device comprises two light source modules **400**, the sensor **700**, the optical protection layer **800**, the light shield **600** and the wire protection layer **210**, wherein the light shield **600** and the wire protection layer **210** are fabricated in a single process.

[0042] It is learned from the above discussion: the image sensing device has the following advantages: [0043] 1. The substrate **100** is used as the carrier to perform the panel-level fabrication process and package process and thus achieve the economic efficiency; [0044] 2. The carrier **500**, which may have different heights, is used to control the relative height of the image module **300** and the light source module **400**; the image module connection member **140** and the light source module connection member **150** are used to accurately control the distance between the image module **300** and the light source module **400**, whereby are avoided the errors that are likely to occur in the conventional assembly operations; [0045] 3. The light shield **600** is filled into the area around the image module **300** and the light source module **400** to effectively prevent stray light from entering the image module **300** and protect the image module **300** and the light source module **400**; [0046] 4. The connection wire **200** is not soldered onto the second surface **120** of the substrate **100** but is soldered onto the first surface **110** where the other elements are also disposed, whereby the size of the image sensing device and the size of the distal tip of the side-viewing endoscope are minimized; [0047] 5. The connection wire **200** is exempted from using an adapter board but is directly soldered onto the substrate **100**, whereby to guarantee the completeness of signals, improve the signal-to-noise ratio, and enhance the image quality; [0048] 6. The sensor **700** is used to monitor the status inside the human body lest the human body be harmed; and [0049] 7. The image sensing device may be equipped with several light source modules **400**, which can emit lights of different wavelengths and colors and can be switched to respectively provide different types of illuminations or simultaneously provide illuminations.

[0050] While the invention is susceptible to various modifications and alternative forms, a specific example thereof has been shown in the drawings and is herein described in detail. It should be understood, however, that the invention is not to be limited to the particular form disclosed, but to the contrary, the invention is to cover all modifications, equivalents, and alternatives falling within the appended claims.

Claims

1. An image sensing device for a side-viewing endoscope, the image sensing device being disposed in a distal tip of the endoscope and comprising: a substrate, including a first surface and a second surface opposite to the first surface, and also including a wire connection member, an image module connection member, and a light source module connection member, which are all arranged on the first surface, wherein the wire connection member is electrically connected with the image module connection member; the wire connection member is also electrically connected with the light source module connection member; an image module, electrically connected with and installed on the image module connection member, and capturing external images along a lateral-

- side direction of the distal tip, wherein the lateral-side direction of the distal tip is a direction parallel to the lateral side of the distal tip and substantially vertical to a first direction; and at least one light source module, electrically connected with and installed on the light source module connection member, the least one light source module being configured to project light along lateral-side direction of the distal tip.
2. The image sensing device for a side-viewing endoscope according to claim 1, wherein the light source module connection member is disposed on the first surface and at positions including either side or both sides of the image module connection member.
 3. The image sensing device for a side-viewing endoscope according to claim 1, further comprising a connection wire, wherein one end of the connection wire is electrically connected with the wire connection member; the wire connection member, the image module connection member, and the light source module connection member are arranged along a first direction; the first direction is parallel to a line from a first end of the first surface to a second end of the first surface, which is opposite to the first end.
 4. The image sensing device for a side-viewing endoscope according to claim 3, wherein the light source module connection member is disposed on the first surface and at one side of the image module connection member, which is corresponding to the first end, along the first direction; the wire connection member is disposed on the first surface and at one side of the image module connection member, which is corresponding to the second end, along the first direction.
 5. The image sensing device for a side-viewing endoscope according to claim 3, wherein the light source module connection member is disposed on the first surface and at one side of the image module connection member, which is corresponding to the second end, along the first direction; the wire connection member is disposed on the first surface and at one side of the image module connection member, which is corresponding to the first end, along the first direction.
 6. The image sensing device for a side-viewing endoscope according to claim 3, wherein the light source module connection member is disposed on the first surface and at two sides of the image module connection member, which are corresponding to the first end and the second end, along the first direction; the wire connection member is disposed on the first surface and at a position between the light source module connection member and the first end; alternatively, the wire connection member is disposed on the first surface and at a position between the light source module connection member and the second end.
 7. The image sensing device for a side-viewing endoscope according to claim 1, further comprising a carrier, which is disposed between the light source module and the light source module connection member.
 8. The image sensing device for a side-viewing endoscope according to claim 7, further comprising an intermediate carrier, which is disposed between the carrier and the light source module.
 9. The image sensing device for a side-viewing endoscope according to claim 1, further comprising a light shield, which are disposed between the light source module and the image module.
 10. The image sensing device for a side-viewing endoscope according to claim 9, wherein the light shield is disposed around the light source module.
 11. The image sensing device for a side-viewing endoscope according to claim 9, wherein the light shield is disposed around the image module.
 12. The image sensing device for a side-viewing endoscope according to claim 9, wherein an optical protection layer is formed on a surface of the light source module.
 13. The image sensing device for a side-viewing endoscope according to claim 1, wherein the light source module is a light-emitting diode or a combination of a fluorescent powder layer and a light-emitting element, which generates a white light source.
 14. The image sensing device for a side-viewing endoscope according to claim 1, wherein the first surface further includes a sensor connection member; the sensor connection member is electrically connected with the wire connection member; a sensor is installed in the sensor connection member.

15. The image sensing device for a side-viewing endoscope according to claim 1, further comprising a connection wire, wherein one end of the connection wire is electrically connected with the wire connection member; a wire protection layer is disposed at a position of the connection wire, which is corresponding to the wire connection member.

16. The image sensing device for a side-viewing endoscope according to claim 1, wherein the light source module includes light-emitting elements for respectively generating lights with different wavelength; the wire connection member a control wire is connected with the light-emitting elements to receive control signals for controlling wavelengths.
